# STAT 3400 - Homework 4

Taketo Horigome

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## Problem 1

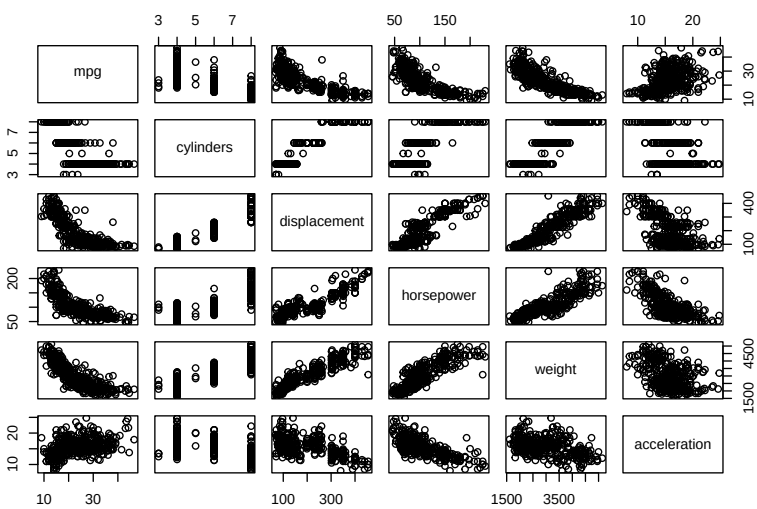
This question involves the use of multiple linear regression on the `Auto` data set from the `ISLR2` library.

### Part A

Produce a matrix of scatter plots that depicts the association between all pairs of numerical variables in the data set. Remove the columns for `year`, `origin`, and `name` prior to constructing your graphic. Describe the shape, direction, and strength of association between all pairs of variables in your graphic.

```r
library(ISLR2)
data(Auto)

pairs(Auto[, -match(c('year', 'origin', 'name'), names(Auto))])
```




**Solution:**

- `mpg` shows a strong negative linear relationship with `horsepower` and `weight`, indicating that heavier cars and cars with more powerful engines tend to have lower fuel efficiency. There is also a moderately strong negative relationship between `mpg` and `displacement`. In contrast, it has a weak to moderate positive relationship with `acceleration`. The relationship with `cylinders` appears weak and non-linear, largely drive by discrete cylinder counts rather than a continuous trend.
- `cylinders` exhibits a positive association with both `displacement` and `weight`, as cars with more cylinders tend to have larger engines and higher mass. However, relationship between `cylinders` and `horsepower` or `acceleration` are less clear and appear weaker due to the discrete nature of the variable.

- `displacement` has a strong positive linear relationship with both `horespoewr` and `weight` suggesting that larger engines produce more power and are typically found in heavier vehicles. Its relationship with `acceleration` is moderately negative, through more dispersed.
- `horsepower` is strongly positively related to `weight`, indicating that heavier cars generally require more powerful engines. It also shows a moderately strong negative relationship with `acceleration`, meaning higher-horsepower vehicles tend to accelerate more quickly.
- `weight` and `acceleration` display a weak negative relationship, with substantial variability, suggesting that weight alone does not strongly determine acceleration. In summary, the strongest relationships occur among `displacement`, `horsepower`, and `weight`, while `mpg` is strongly negatively associated with these variables.

## Part B

Estimate a multiple linear regression model to explain a vehicle's gas mileage (`mpg`) based on the other numerical variables in the data set. Remove the columns for `year`, `origin`, and `name` prior to fitting your model. Based on the summary output, interpret the model's residual standard error (RSE), coefficient of determination (R-squared), and $F$-statistic in the context of the problem.

```r
range(Auto$mpg)
```

```
## [1]  9.0 46.6
```

```r
model <- lm(mpg ~ ., data = subset(Auto, select = -c(year, origin, name)))
summary(model)
```

```
##
## Call:
## lm(formula = mpg ~ ., data = subset(Auto, select = -c(year, origin,
##     name)))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.5816  -2.8618  -0.3404   2.2438  16.3416
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.626e+01  2.669e+00  17.331   <2e-16 ***
## cylinders    -3.979e-01  4.105e-01  -0.969   0.3330
## displacement -8.313e-05  9.072e-03  -0.009   0.9927
## horsepower   -4.526e-02  1.666e-02  -2.716   0.0069 **
## weight       -5.187e-03  8.167e-04  -6.351    6e-10 ***
## acceleration -2.910e-02  1.258e-01  -0.231   0.8171
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.247 on 386 degrees of freedom
## Multiple R-squared:  0.7077, Adjusted R-squared:  0.7039
## F-statistic: 186.9 on 5 and 386 DF,  p-value: < 2.2e-16
```

**Solution:** The residual standard error is 4.247 mpg indicating that the model's predicted fuel efficiency values differ from the observed values by about 4.25 mpg on average. Given that `mpg` ranges from approximately 9 to 46.6, this suggests reasonably accurate predictions. The multiple R-squared value of 0.7077 indicates that about 70.8 of the variability in fuel efficiency is explained by hte predictors in the model. The adjusted R-squared %0.7039% is slightly lower and accounts for the number of predictors, making it useful for model comparison. The F-statistic of 186.9 with a p-values less than 0.05 provides strong evidence that at least

one predictor is significantly associated with `mpg`, indicating that the model is statistically significant overall. Using a significance level of 0.05, only `horsepower` and `weight` are statistically significant predictors of `mpg`, both with negative coefficients, while the remaining predictors are not significant after accounting for the others.

# Problem 2

This problem references the model you constructed in Part B of the previous problem. Reproduce the summary output for the model and review the `Coefficients` table. Describe the null hypotheses to which the $p$-values given in this table correspond. Explain what conclusions you can draw based on these $p$-values. Your explanation should be phrased in terms of a vehicle's performance characteristics (e.g., horsepower and cylinders), rather than in terms of the coefficients of the model.

```r
summary(model)
```

```
##
## Call:
## lm(formula = mpg ~ ., data = subset(Auto, select = -c(year, origin,
##     name)))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.5816  -2.8618  -0.3404   2.2438  16.3416
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.626e+01  2.669e+00  17.331   <2e-16 ***
## cylinders    -3.979e-01  4.105e-01  -0.969   0.3330
## displacement -8.313e-05  9.072e-03  -0.009   0.9927
## horsepower   -4.526e-02  1.666e-02  -2.716   0.0069 **
## weight       -5.187e-03  8.167e-04  -6.351    6e-10 ***
## acceleration -2.910e-02  1.258e-01  -0.231   0.8171
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.247 on 386 degrees of freedom
## Multiple R-squared:  0.7077, Adjusted R-squared:  0.7039
## F-statistic: 186.9 on 5 and 386 DF,  p-value: < 2.2e-16
```

**Solution:** From the results, `horsepoewr` has p-value of 0.0069, which is smaller than the significance level of 0.05. This means `horsepower` is an important performance characteristic for predicting `mpg`. In general, cars with higher `horsepower` tend to have lower fuel efficiency. In addition, `weight` has smaller p-value than the alpha, which means `weight` is also strongly related to `mpg`. Heavier vehicles tend to get worse gas mileage. On the other hand, `cylinders`, `displacement` and `acceleration` all have p-values larger than 0.05. Therefore, we fail to reject the null hypotheses for these variables, which suggests that after accounting for `horsepower` and `weight`, these characteristics do not provide much additional information for predicting `mpg`. Lastly, the p-value for the overall F-test is less than 0.05, which means the model is statistically significant overall. This tells us that at least one vehicle characteristic in the model is realted to `mpg`.

# Problem 3

This problem references the model you constructed in Part B of Problem 1. Using the same model, compute a 95% confidence interval and 95% prediction interval for the fuel efficiency (`mpg`) of a car with the following

characteristics: `cylinders`=4, `displacement`=92, `horsepower`=85, `weight`=2200, and `acceleration`=17. Interpret each of your intervals in the context of the problem.

```
new_car <- data.frame(cylinders = 4, displacement = 92, horsepower = 85, weight = 2200, acceleration =

predict(model, newdata = new_car, interval = 'confidence', level = 0.95)
```

```
##       fit     lwr      upr
## 1 28.9121 28.1462 29.67799
```

**Solution:** We are 95% confident that the fuel efficiency of a single car with these characteristics will fall between 28.15 and 29.68 mpg.

# Problem 4

This problem involves the `Boston` data set from the `ISLR2` library. We will now try to explain per capita crime rate using the other variables in this data set. In other words, per capita crime rate is the response, and the other variables are the predictors.

## Part A

For each predictor, fit a simple linear regression model to explain the response. Describe your results. In which of the models is there a statistically significant association between the predictor and the response? Which predictor offers the best fit based on either RSE or R-squared?

```
data(Boston)

for (col in names(Boston)[-1]) {
  cat("\nPredictor:", col, "\n")
  print(summary(lm(as.formula(paste("crim ~", col)), data = Boston)))
}
```

```
##
## Predictor: zn
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -4.429 -4.222 -2.620  1.250 84.523
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.45369    0.41722  10.675  < 2e-16 ***
## zn          -0.07393    0.01609  -4.594 5.51e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.435 on 504 degrees of freedom
## Multiple R-squared:  0.04019,    Adjusted R-squared:  0.03828
## F-statistic:  21.1 on 1 and 504 DF,  p-value: 5.506e-06
##
##
## Predictor: indus
```

```
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.972  -2.698  -0.736   0.712  81.813
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.06374    0.66723  -3.093  0.00209 **
## indus        0.50978    0.05102   9.991  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.866 on 504 degrees of freedom
## Multiple R-squared:  0.1653, Adjusted R-squared:  0.1637
## F-statistic: 99.82 on 1 and 504 DF,  p-value: < 2.2e-16
##
##
## Predictor: chas
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -3.738 -3.661 -3.435  0.018 85.232
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.7444     0.3961   9.453   <2e-16 ***
## chas         -1.8928     1.5061  -1.257    0.209
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.597 on 504 degrees of freedom
## Multiple R-squared:  0.003124,   Adjusted R-squared:  0.001146
## F-statistic: 1.579 on 1 and 504 DF,  p-value: 0.2094
##
##
## Predictor: nox
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.371  -2.738  -0.974   0.559  81.728
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -13.720      1.699  -8.073 5.08e-15 ***
## nox           31.249      2.999  10.419  < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 504 degrees of freedom
## Multiple R-squared:  0.1772, Adjusted R-squared:  0.1756
## F-statistic: 108.6 on 1 and 504 DF,  p-value: < 2.2e-16
##
##
## Predictor: rm
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.604 -3.952 -2.654  0.989 87.197
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   20.482      3.365   6.088 2.27e-09 ***
## rm            -2.684      0.532  -5.045 6.35e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.401 on 504 degrees of freedom
## Multiple R-squared:  0.04807,    Adjusted R-squared:  0.04618
## F-statistic: 25.45 on 1 and 504 DF,  p-value: 6.347e-07
##
##
## Predictor: age
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.789 -4.257 -1.230  1.527 82.849
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.77791    0.94398  -4.002 7.22e-05 ***
## age          0.10779    0.01274   8.463 2.85e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.057 on 504 degrees of freedom
## Multiple R-squared:  0.1244, Adjusted R-squared:  0.1227
## F-statistic: 71.62 on 1 and 504 DF,  p-value: 2.855e-16
##
##
## Predictor: dis
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
```

```
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.708 -4.134 -1.527  1.516 81.674
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.4993     0.7304  13.006   <2e-16 ***
## dis          -1.5509     0.1683  -9.213   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.965 on 504 degrees of freedom
## Multiple R-squared:  0.1441, Adjusted R-squared:  0.1425
## F-statistic: 84.89 on 1 and 504 DF,  p-value: < 2.2e-16
##
##
## Predictor: rad
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.164  -1.381  -0.141   0.660  76.433
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.28716    0.44348  -5.157 3.61e-07 ***
## rad          0.61791    0.03433  17.998  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.718 on 504 degrees of freedom
## Multiple R-squared:  0.3913, Adjusted R-squared:   0.39
## F-statistic: 323.9 on 1 and 504 DF,  p-value: < 2.2e-16
##
##
## Predictor: tax
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.513  -2.738  -0.194   1.065  77.696
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.528369   0.815809  -10.45   <2e-16 ***
## tax          0.029742   0.001847   16.10   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 6.997 on 504 degrees of freedom
## Multiple R-squared:  0.3396, Adjusted R-squared:  0.3383
## F-statistic: 259.2 on 1 and 504 DF,  p-value: < 2.2e-16
##
##
## Predictor: ptratio
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -7.654 -3.985 -1.912  1.825 83.353
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -17.6469     3.1473  -5.607 3.40e-08 ***
## ptratio       1.1520     0.1694   6.801 2.94e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.24 on 504 degrees of freedom
## Multiple R-squared:  0.08407,    Adjusted R-squared:  0.08225
## F-statistic: 46.26 on 1 and 504 DF,  p-value: 2.943e-11
##
##
## Predictor: lstat
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -13.925  -2.822  -0.664   1.079  82.862
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.33054    0.69376  -4.801 2.09e-06 ***
## lstat        0.54880    0.04776  11.491  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.664 on 504 degrees of freedom
## Multiple R-squared:  0.2076, Adjusted R-squared:  0.206
## F-statistic:   132 on 1 and 504 DF,  p-value: < 2.2e-16
##
##
## Predictor: medv
##
## Call:
## lm(formula = as.formula(paste("crim ~", col)), data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
```

```
## -9.071 -4.022 -2.343  1.298 80.957
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 11.79654    0.93419   12.63   <2e-16 ***
## medv        -0.36316    0.03839   -9.46   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.934 on 504 degrees of freedom
## Multiple R-squared:  0.1508, Adjusted R-squared:  0.1491
## F-statistic: 89.49 on 1 and 504 DF,  p-value: < 2.2e-16
```

**Solution:** Based on the results, most predictors show a statistically significant association with crime, including `zn`, `indus`, `nox`, `rm`, `age`, `dis`, `rad`, `tax`, `ptratio`, `lstat`, and `medv`, since their p-values are smaller than 0.05, The only predictor that does not show a statistically significant association is `chas`. The only variable that does not show a statistically significant association with crime is `chas` (Charles River dummy variable), since its p-value is greater than 0.05.

## Part B

Fit a multiple regression model to explain the response using *all* of the predictors. Describe your results. For which predictors can we reject the null hypothesis: $H_0 : \beta_j = 0$?

```r
multi_model <- lm(crim ~., data = Boston)
summary(multi_model)
```

```
##
## Call:
## lm(formula = crim ~ ., data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.534 -2.248 -0.348  1.087 73.923
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.7783938  7.0818258   1.946 0.052271 .
## zn           0.0457100  0.0187903   2.433 0.015344 *
## indus       -0.0583501  0.0836351  -0.698 0.485709
## chas        -0.8253776  1.1833963  -0.697 0.485841
## nox         -9.9575865  5.2898242  -1.882 0.060370 .
## rm           0.6289107  0.6070924   1.036 0.300738
## age         -0.0008483  0.0179482  -0.047 0.962323
## dis         -1.0122467  0.2824676  -3.584 0.000373 ***
## rad          0.6124653  0.0875358   6.997 8.59e-12 ***
## tax         -0.0037756  0.0051723  -0.730 0.465757
## ptratio     -0.3040728  0.1863598  -1.632 0.103393
## lstat        0.1388006  0.0757213   1.833 0.067398 .
## medv        -0.2200564  0.0598240  -3.678 0.000261 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.46 on 493 degrees of freedom
## Multiple R-squared:  0.4493, Adjusted R-squared:  0.4359
```

```
## F-statistic: 33.52 on 12 and 493 DF,  p-value: < 2.2e-16
```

**Solution:** In the multiple regression model, several predictors are statistically significant. In particular, `zn`, `dis`, `rad`, and `medv` have p-values smaller than 0.05, so we reject the null hypothesis that their coefficients are zero. This suggests taht these variables are associated with per capita crime rate, `crim`, even after accounting for the other predictors in the model. The overall F-test has a very small p-value, indicating that model is statistically significant.

## Part C

How do your coefficient estimates from Part A compare to those from Part B? Create a plot displaying the simple linear regression coefficients from Part A on the $x$-axis, and the multiple linear regression coefficients from Part B on the $y$-axis. That is, each predictor is displayed as a single point in the plot. Its coefficient in a simple linear regression model is shown on the $x$-axis, and its coefficient estimate in the multiple linear regression model is shown on the $y$-axis. Interpret your plot in the context of the problem.

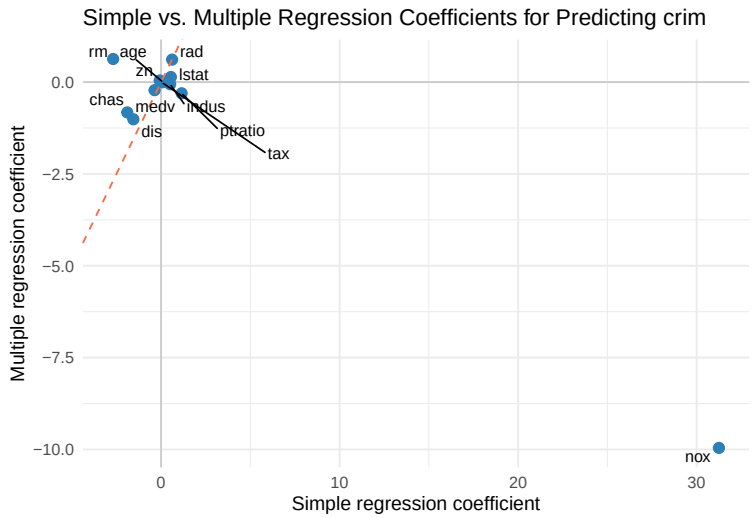
```r
library(ggrepel)
simple_df <-
  names(Boston)[names(Boston) != "crim"] %>%
  map_df(function(var) {
    fit <- lm(reformulate(var, response = "crim"), data = Boston)
    tibble(
      predictor = var,
      coef_simple = coef(fit)[[2]]
    )
  })

multi_fit <- lm(crim ~ ., data = Boston)

multi_df <-
  tibble(
    predictor = names(coef(multi_fit))[-1],
    coef_multiple = unname(coef(multi_fit))[-1]
  )

coef_df <- simple_df %>%
  inner_join(multi_df, by = "predictor")

ggplot(coef_df, aes(x = coef_simple, y = coef_multiple, label = predictor)) +
  geom_hline(yintercept = 0, color = "grey80") +
  geom_vline(xintercept = 0, color = "grey80") +
  geom_point(color = "#2C7FB8", size = 2.8) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "#FB6A4A") +
  geom_text_repel(size = 3.5, max.overlaps = Inf) +
  labs(
    title = "Simple vs. Multiple Regression Coefficients for Predicting crim",
    x = "Simple regression coefficient",
    y = "Multiple regression coefficient"
  ) +
  theme_minimal(base_size = 12)
```




**Solution:** Most predictors fall close to the dashed $y = x$ line, which means their simple-regression slopes and multiple-regression slopes are very similar. However, variables like `nox`, `rm`, and `ptratio` sit far from the line, showing that their effects change a lot once we control for the other predictors. This suggests these variables are more heavily influenced by multicollinearity or shared variation with the rest of the dataset.